



Lab Overview

A. Lab Overview

There are 13 labs starting from Week 1. Among these, 9 labs are structured around core teaching topics, focusing on key technical and practical skills. The remaining 4 labs are allocated for project demonstrations, assignment assessment, and milestone reviews, providing dedicated time for student presentations and tutor feedback.

There are **eight individual tasks** designed to engage students with the core lab content and ensure that each team member is actively contributing to the group project. Each individual task, along with parts of the assignments, will be **marked in-lab (on-site, face-to-face)**. You must be present in person to receive marks.

INFO

Lab and assignment content is provided through this website, with the relevant links posted on Canvas. Once a content link is posted or updated on Canvas, the linked version is considered final and confirmed. Check Canvas regularly for new or updated links.

LAB ATTENDANCE AND GROUP FORMATION

You must attend the **lab session you're officially enrolled in**, based on your allocated lab code in the university system.

Attending a different lab without formal approval does **not** allow you to form or join a group in that session.

If you have a timetable issue, please contact the **unit coordinator or tutors** as soon as possible.

WEEK 1: NO ON-SITE LAB

Due to university scheduling, there is no on-site lab in **Week 1**. Students are still **required** to read the **Week 1 Lab** content and complete **Individual Task 1** as normal.

A.1 Lab Outline

Week	Lab Topic (3hr)
1	01 Structured Planning and System Design for Web Applications
2	02 RESTful API Design and Team Collaboration with GitHub
3	03 Start Your First Django Web App
4	04 Django ORM, Django Model and Django Admin
5	A1 Milestone Presentation
6	05 Django Authentication and File Storage System
7	06 DRF and Django Advance
8	07 Security and Testing in Django
9	08 AWS Cloud & GraphQL
10	09 Deploy Your Project
11	Project Help & Review Session
12	A2 Tech Interview
13	A3 Tech Interview

A.2 Knowledge Checks

Knowledge Checks are **optional interactive review materials** designed to strengthen your understanding of important lab concepts and help you identify topics that may need further revision. They are not assessment tasks and do not contribute to your marks.

Your answers and results are **not submitted to or saved by the teaching team**. The activities run locally in your browser using JavaScript. A small completion record may be kept only in

your browser's local storage for convenience; it is not uploaded, monitored, or used for assessment and may disappear if you clear your browser data or use another device.

! AI-ASSISTED CONTENT

The questions and interactive explanations were created with **AI assistance**. Although they have been reviewed, AI-assisted content may still contain mistakes, ambiguity, or oversimplifications. If an answer seems unclear or incorrect, or you would like to discuss the reasoning, please ask on [Ed Discussion](#).

B. Lab Activity

Each lab is primarily hands-on and self-guided. Students are expected to:

- Read and understand the lab content independently.
- Complete the associated individual task for the lab.
- Attend in-person labs to have their previous task marked.
- Engage in Q&A with tutors regarding the current lab or ongoing project work.

Tutors will provide weekly updates, mark individual submissions in-lab, and address student questions.

B.1 Pre-lab Preparation

Before attending each lab, you should:

1. **Review the previous lab content thoroughly.**
2. **Complete and submit the previous week's individual task on Canvas before the deadline.**
3. **(Optional)** Read the current lab guide in advance so that you can ask specific questions during the lab.

B.2 Lab Timeline

Time	Activity
0:10 - 0:30	Tutors will make announcements (e.g., assignment releases), summarise the key points from the previous lab, and introduce the next individual task , highlighting important expectations and potential pitfalls.
0:30 - 2:00	Tutors will mark each student's individual task submission from the previous lab. Each student will get around 4 minutes to explain or demo their work. In the meantime, you can either work on the individual task, read each week's lab guide, or discuss your group project.
2:00 - 3:00	Remaining time is open for questions and consultation , including current lab tasks, project progress, assignment preparation, or technical support.

B.3 Individual Task

There are **eight individual tasks (Lab 01–08)**, each worth 1%. Each task is:

- Released together with the corresponding lab.
- Due **by the following Wednesday midnight** (e.g., Lab 1 task is due Week 2 Wednesday night).

SUBMISSION DEADLINE 🤖🤖🤖

 **Exception:** The Lab 4 task (Week 4) is due in **Week 6**, because Week 5 is allocated for the A1 Milestone Presentation.

 If you're unsure about a deadline, always refer to the **Canvas assignment deadline**, which is considered final.

What to Submit

- Submit a **single PDF** report via Canvas.
- The format is flexible but must **follow the task guideline** provided in the lab.
- For technical tasks (e.g., coding, diagrams), it's acceptable to include **screenshots** in your report.

HOW MUCH DO I NEED TO WRITE?

There is no fixed word or page count, and you do not need to write more than necessary. Your report should clearly address the required discussion and provide sufficient evidence, such as code snippets, diagrams, or screenshots. **Five pages should be quite enough.** The PDF supports what you explain during the lab; it does not replace the in-person explanation or demonstration.

How We Mark

- Tutors will mark your work **in-person during the lab** in the **week the task is due**.
- You must **present, explain, or demo** your work (as documented in your report), and there may be some short **Q&A** during the process.
- While Canvas submission is required and will be reviewed as supporting evidence, **marks are awarded based on your live explanation/demo** during the lab.

BE READY FOR MARKING

If your report includes screenshots or explanations but you fail to cover those in your live demo, **you may still lose marks.**

If your explanation during the lab has **significant inconsistencies** with your submitted report, you will also **lose marks.**

Prepare your report, notes, or demo project ahead of time to support your in-lab explanation.

Summary

Deliverable	Format	Submission Platform	Notes
Report	PDF	Canvas	Submit before the deadline; used as proof of work.
In-Lab Marking	N/A	In Lab	Marking is based on what you show or explain.

LATE SUBMISSION

In accordance with University policy, these penalties apply when written work is submitted after 11:59pm on the due date:

- Deduction of **5%** of the maximum mark for each calendar day after the due date.
- After ten calendar days late, a mark of zero will be awarded.

C. AWS Academy

All students have been invited to join **two AWS Academy courses** and **AWS Learner Lab** through the AWS Academy platform. AWS Learner Lab environment will be used later in the semester for **your deployment task in the final assignment**, so it's essential that **you ensure access as early as possible**.

While these two AWS Academy courses are **not mandatory** and will **not be examined** in the final exam, they are highly recommended. They offer a **comprehensive and structured introduction to cloud computing architecture**, which will help you better understand how internet platforms are built and deployed. Upon completion, you will also receive **two official AWS Academy Certificates**.

Please check your **university email** for invitations to the AWS Academy resources. Use the **email that received the invitation** (typically your Unikey email) to register.



TIP

If you have **never used AWS Academy before**, you must first create an **AWS Academy (Canvas)** account — note this is **not the same** as your USYD Canvas. See the following instructions on how to access AWS Academy resources, using the AWS Learner Lab as an example.

C.1 Provided AWS Academy Courses

1. **AWS Academy Cloud Foundations**

- Covers core AWS services and concepts, such as compute, storage, and networking.
- Ideal for beginners looking to build foundational knowledge of cloud platforms.

2. **AWS Academy Cloud Architecting**

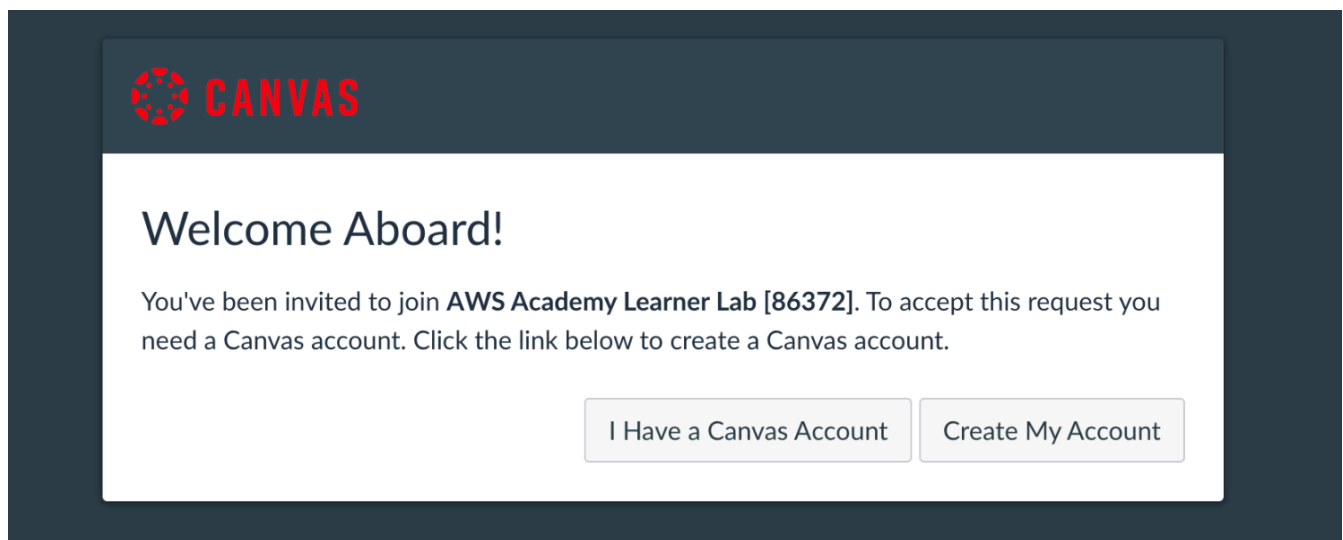
- Dives deeper into scalable and secure cloud solution design.
- Aligned with the AWS Certified Solutions Architect – Associate certification.

C.2 How to Access the AWS Learner Lab

Two documents are provided on **Canvas** to guide you:

- **AWS Academy Learner Lab – Student Guide**
- **AWS Academy Learner Lab – README** (usage rules and expectations)

When accessing AWS Academy for the first time via an invitation link, you will be prompted to the page as below. You can either sign in using your existing AWS Academy Canvas account **(not your USYD Canvas Account !!)** or create a new account by selecting "Create My Account." When creating a new account, you'll only need to set a password without providing any additional information.



Follow the steps in the Student Guide carefully. If you **have not received an invitation** or encounter **any access issues**, please ask your **tutor for help during the lab**.

DANGER

Please note, you should not be asked to provide any personal information, such as your university account password or credit card details. If you encounter such requests, contact tutors immediately.